Environmental Engineering

Unit IV: Planning and Design of the Sewerage System

Syllabus

Characteristics and composition of sewage - Population equivalent - Sanitary sewage flow estimation - Sewer materials - Hydraulics of flow in sanitary sewers - Sewer design - Storm drainage - Storm runoff estimation - Sewer appurtenances - Corrosion in sewers - Prevention and control - Sewage pumping - Drainage in buildings - Plumbing systems for drainage.

1. Introduction to Sewerage

Sewage is the liquid waste from a community: domestic, industrial and storm water (with infiltration). **Sewerage** is the system of collecting, conveying, treating and disposing of it. **Sanitation** is the whole job of keeping the community clean and healthy.

Terms to know

Term	Meaning
Sewage / wastewater	Liquid waste from houses, institutions, industries
Sullage	Wastewater from bathrooms, kitchens, washing (without human excreta)
Sanitary sewage	Domestic + industrial sewage (with excreta)
Storm water	Run-off from rainfall
Sewer	Underground pipe carrying sewage
Sewerage	Collection system (sewers) with appurtenances
Night soil	Human excreta
Refuse / garbage	Solid waste

Systems of sewerage

System	Description	Merits / Demerits
Separate system	Sanitary sewage and storm water in **separate** sewers	Smaller sewers for sewage, cheaper treatment (no storm water), no overflow of sewage; **two sets of pipes cost more**, storm sewer lies idle in dry weather
Combined system	One sewer carries both	One pipe only, cheaper where rainfall is uniform, self-cleansing in rain; **large sewers, treatment overloaded in rains, overflow** problem
Partially separate (partially combined)	Part of storm water (roof and courtyards) goes into the sanitary sewer	Compromise, **good cleansing**

- Separate system is **preferred in India** (heavy monsoon rainfall in short duration)
- Combined system is suited where rainfall is evenly spread, or in areas with dense population and no space for two pipes

2. Characteristics and Composition of Sewage

Fresh sewage is **grey, turbid** with slight odour; **stale or septic** sewage is dark, with a foul smell of H_2S. Composition: **99.9% water** and only **0.1% solids** (suspended + dissolved). Of the solids about 70% is organic, 30% inorganic.

Physical characteristics

- **Temperature** - slightly higher than water supply (20-30 degrees C), affects biological activity
- **Colour** - grey (fresh), black (septic); **odour** - from H_2S, indole, skatole
- **Turbidity** - due to suspended and colloidal solids
- **Solids** - Total (TS), Suspended (SS), Dissolved (DS), **volatile (organic)** and **fixed (inorganic)**; settleable solids measured by Imhoff cone

Chemical characteristics

Parameter	Meaning / typical value (domestic sewage)
BOD_5	Oxygen needed by bacteria to oxidise organic matter in 5 days at 20 degrees C; **200 – 300 mg/L**
COD	Oxygen to oxidise organic matter chemically; 400 – 600 mg/L; COD > BOD
Dissolved oxygen (DO)	Nil in septic sewage; the discharge must keep DO > 4 mg/L in stream
Total suspended solids	200 – 400 mg/L
Nitrogen	Organic N, ammonia, nitrite, nitrate; TKN 40-60 mg/L; ammonia indicates fresh pollution
Phosphorus	5 – 15 mg/L, causes **eutrophication**
pH	6.5 – 8; **chlorides** 100 – 200 mg/L; **alkalinity** 100-200 mg/L
Oil and grease	Reduce oxygen transfer; remove in grease traps
Heavy metals and toxic compounds	From industries (Pb, Cr, Cd, Hg), toxic to bacteria

Biological characteristics

- Bacteria (aerobic, anaerobic, facultative) – **billions per mL**; viruses, protozoa, helminth eggs, algae, fungi
- **Coliform test (MPN)** indicates faecal contamination
- Bacteria decompose organic matter – the basis of **biological treatment**

BOD equation

$BOD_t = L_0 (1 - e^{-kt})$ [first-order reaction]

L_0 = ultimate BOD; k = BOD rate constant (0.23 per day at 20 degrees C for sewage, base e = 0.1 per day base 10)

Temperature correction: $k_T = k_{20} \times (1.047)^{T-20}$; BOD_5 is about 68% of ultimate BOD

3. Population Equivalent

To express the strength of industrial wastewater in terms of domestic sewage, the **population equivalent (PE)** is used. It is the population which would give the same BOD load as the industrial waste.

Population equivalent

PE = (BOD_5 of waste in kg/day) / (BOD per capita per day)

Per capita BOD load = **0.08 kg/day (80 g/day)** (some books use 0.045-0.085)

Also PE = (flow of waste in L/day x BOD mg/L) / (80 g/day), with units adjusted

Example: A dairy releases 500 m^3/day with BOD 600 mg/L => 300 kg/day => PE = 300/0.08 = **3750**

4. Estimation of Sanitary Sewage Flow

- About **80%** of water supplied becomes sewage (the rest is lost in lawn sprinkling, evaporation, leaks)
- **Average dry weather flow (DWF)** = 80% x per capita water supply x population = 0.8 x 135 = **108 lpcd**
- Add **infiltration** of ground water through joints: 5-10% of DWF (or 0.0005 to 0.005 m^3/s/km, or 1500-2000 L/day/km/cm diameter)
- **Industrial waste** added separately; **public places** also included

Flow	Multiplying factor
Average DWF	1
Maximum flow (peak)	**2 to 3 times** the average (depends on population)
Minimum flow	1/3 to 1/2 of the average (for population over 1 lakh, up to 1/2)
Design flow for sewers	**Peak flow**, 2.25 - 3 x average

Peak factor (CPHEEO)

Population up to 20,000 : 3.0; 20,000 - 50,000 : 2.5; 50,000 - 7.5 lakh : 2.25; above 7.5 lakh : 2.0

Or Harmon's formula: M = 1 + 14/(4 + sqrt(P)), P in thousands

5. Sewer Materials

Material	Merits	Demerits / use
Stoneware (vitrified clay)	Resists corrosion, smooth, cheap, small sizes, long life	Brittle, difficult to join; up to 300 mm
Cast iron / ductile iron	Strong, tight, for pumping mains, under bridges, high pressure	Costly, corrodes in acidic sewage
Concrete (plain, RCC, PSC)	Large diameter, any shape, strong, durable	Attacked by H_2S acid, heavy, joints; plain up to 600 mm
Brick masonry	Large trunk and storm sewers; any shape	Rough, porous, high friction
Asbestos cement	Light, long length	Not used now
Plastic (uPVC, HDPE, GRP)	Light, **smooth, corrosion free**, quick, leak-proof	Low strength, damage by heat; small-medium sizes
Steel	Pressure mains, bridges	Corrosion

- **Shapes of sewers:** circular (most common, max hydraulic efficiency, easy to make), egg-shaped (good self-cleansing at low flows), **horse-shoe / semi-elliptical / rectangular / U-shaped** (large trunk and storm drains)

6. Hydraulics of Flow in Sanitary Sewers

Sewers are designed to **flow partly full** under gravity (open channel flow) and carry maximum flow at depth near 0.8D. Flow in sewers follows Manning's, Chezy's or Hazen-Williams formula.

Flow formulae

Manning: $V = (1/n)\ R^{2/3}\ S^{1/2}$; $Q = A\ V$ (n = 0.012 - 0.015 for sewers)

Chezy: $V = C\ \sqrt{RS}$; **Kutter** and **Crimp & Bruge:** $V = 83.5\ R^{2/3}\ S^{1/2}$

Full flow circular pipe: $R = D/4$, so $V_{full} = (0.397/n)\ D^{2/3}\ S^{1/2}$

Self-cleansing velocity

- Velocity at which **sediments (grit) are not deposited**, and sewage solids are carried; **minimum 0.6 m/s** at the peak (some codes 0.8 m/s at design flow); **maximum 3 m/s** (non-scouring, avoid abrasion of the sewer); typical 0.8 - 2.4 m/s
- Shield's / Camp formula: $V = (1/n)\ R^{1/6}\ \sqrt{8\ \beta\ (S_s - 1)\ d / f}$ where β = 0.04-0.8, d = grit diameter

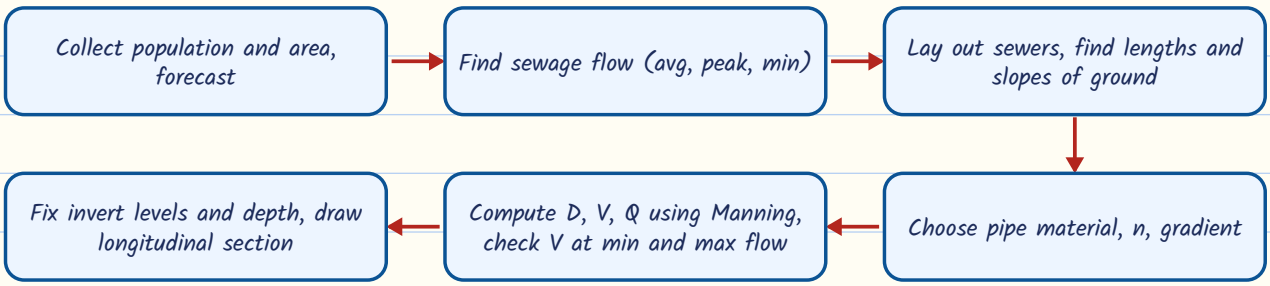
Partial flow in circular sewers (proportionate depth)

Depth ratio d/D	Q / Q_{full}	V / V_{full}
0.25	0.15	0.77
0.50	0.50	1.00
0.80	0.98	1.14 (max V at 0.81D)
0.94	1.07 (max Q near 0.94D)	1.10
1.00	1.00	1.00

Key points

* Max velocity occurs at 0.81 D, max discharge at 0.94 D
* Sewers are designed for 0.5 - 0.8 D at design flow, to leave room for gases and extra flow
* Velocity at minimum flow should be above 0.45 m/s (self-cleansing check)

7. Design of Sanitary Sewers



- **Design period** 30 years; sewers designed for **maximum flow running 0.8 full** (diameter D for peak with d/D = 0.8)
- **Minimum diameter:** 150 mm for public sewers (200 mm for CPHEEO in some cases), 100 mm for house connections
- **Minimum depth of cover:** 1 - 1.2 m (to protect from loads, avoid freezing, serve basements); **maximum depth** 6 m or less
- **Manholes** at changes of direction, gradient, size, junction, and every **30 m (up to 300 mm)**, 60 m for larger
- **Invert level** is the lowest point of the inside of the pipe; the flow line must keep downstream **crown** not above upstream crown **(match crown or match 0.8 depth lines)**

Example

Population 40,000 at 135 lpcd. Sewage = 0.8 x 135 x 40,000 = 4.32 MLD = 0.05 m^3/s. Peak factor 2.5 => Q = 0.125 m^3/s.

Slope 1 in 500 (S = 0.002), n = 0.013, running 0.8 full: Q_{full} = 0.125/0.98 = 0.1276 m^3/s.
Q_{full} = (0.312/n) $D^{8/3}$ $S^{1/2}$ => $D^{8/3}$ = 0.1276 x 0.013 / (0.312 x 0.0447) = 0.119 => D = 0.45 m; provide **450 mm pipe**. Check V_{full} = Q/A = 0.1276/0.159 = 0.80 m/s.

8. Storm Drainage and Storm Runoff Estimation

Storm drains carry the rain water from roofs, roads and open areas to natural streams, to prevent flooding. The design depends on **rainfall intensity, area, and the surface (imperviousness).**

Rational method

Q = (1/360) C i A (Q in m^3/s, i in mm/h, A in hectares)

or Q = C i A with i in m/s, A in m^2

C = run-off coefficient (impervious 0.8 - 0.95, roads 0.7-0.9, lawns 0.05-0.2, parks 0.1-0.25, residential 0.3-0.5)

- **Time of concentration (t_c) = inlet time** (overland flow into the first inlet, 5-20 min) + **time of flow** in the drain (length / velocity); the duration of the design storm equals t_c
- **Intensity-duration-frequency (IDF) formula:** i = a / (t + b)^n or i = K T^x / $(t + a)^n$; design **return period 2-10 years** for towns, up to 25-50 for important areas
- Methods for large areas: **hydrograph methods, Unit hydrograph, Kuichling, Burkli-Ziegler (i = ...), Mc Math**, SCS-CN method, computer models (SWMM)
- **Weighted C** for composite area: C = sum(C_i A_i) / sum(A_i)

Example

A 20 ha area: 40% roads and roofs (C = 0.85), 60% lawns (C = 0.15). Weighted C = 0.4 x 0.85 + 0.6 x 0.15 = 0.43.

t_c = 20 min, intensity 60 mm/h. Q = 0.43 x 60 x 20 / 360 = **1.43 m^3/s**

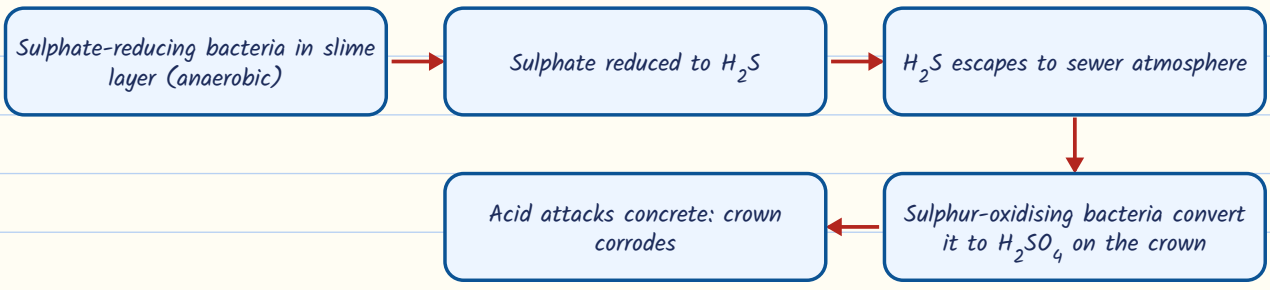
- Storm sewers are usually **open drains** along roads (lined, rectangular, trapezoidal or U-shaped), or **closed** large concrete pipes and box culverts
- **Inlets:** kerb inlets, gutter inlets, **catch basins (pits)** at 30-60 m to trap grit; **storm water overflow** weirs in combined system; **detention ponds, rain-water harvesting** to reduce the peak flow

9. Sewer Appurtenances

Appurtenance	Purpose
Manhole	Access for inspection and cleaning; at junctions, bends, change of size, every 30–90 m. Types: shallow (0.9 m), normal (1.5 m), **deep (above 1.5 m)** with steps and drop manhole. Cover CI or RCC
Drop manhole	Brings a high branch sewer down to a lower main, preventing splashing and erosion
Lamp hole	Straight sewers up to 300 mm, a lamp lowered to check alignment
Clean-out	Pipe up to the surface at the upper ends of sewers for flushing
Catch basin / catch pit	Retain grit, debris from the street inlets
Inlets (street, gully traps)	Admit surface storm water; grating prevents debris entering
Flushing tanks	Flush sewers at dead ends, avoids silting; automatic siphon
Grease and oil traps	Hotels, garages: remove grease
Inverted siphon (depressed sewer)	Crossing under a stream, road or railway; sewer dips below, runs under pressure; velocity above 1 m/s at the minimum flow
Storm overflow (regulator)	In combined system, to divert excess storm flow
Ventilating shafts	Remove sewer gases
Sewer ventilation	Prevents explosion and corrosion; via manhole covers, vent pipes

10. Corrosion in Sewers: Prevention and Control

Corrosion of concrete (and metal) sewers is chiefly due to **hydrogen sulphide (H_2S) formed in anaerobic septic sewage**.



Reactions

SO_4^{2-} + organic matter -> S^{2-} + CO_2 + H_2O (sulphate-reducing bacteria)

$H_2S + 2O_2 \rightarrow H_2SO_4$ (Thiobacillus, above the water line)

$H_2SO_4 + CaCO_3$ (cement) -> $CaSO_4 + H_2O + CO_2$ (expands, cracks, spalls)

- **Other problems of H_2S:** foul odour (rotten egg smell), toxic gas (**dangerous to maintenance workers**), blackening of paints, explosion of gases (methane)
- **Prevention and control:** keep sewage fresh and **aerobic**: keep velocity above 0.6 m/s, ventilation, avoid long detention, **air or oxygen injection** into force mains, steep gradients
- **Chemical dosing:** chlorine, hydrogen peroxide, ferrous / ferric salts (precipitate sulphide), lime (raise pH above 9), nitrate, ozone
- **Use of resistant materials:** vitrified clay, HDPE / PVC, **epoxy / PVC lining on concrete**, high alumina cement, additional sacrificial cover, **coat with bitumen**
- Control industrial discharges (acidic waste, high sulphate); regular inspection and cleaning of slime

11. Sewage Pumping

Pumping of sewage is needed when the ground is **flat**, sewers become too deep, or low-lying areas are to be drained, or the effluent must be lifted to treatment works.

- **Pumping stations:** wet well (receives sewage, with **screens** before it) + dry well (pumps and motors) or **submersible pumps** in the wet well
- Pump types: **centrifugal non-clog pumps** with open or semi-open impeller, **screw pumps** (Archimedean) for large flows at low heads, **pneumatic ejectors**, submersible pumps
- **Wet well detention time** 10-30 min (to avoid septicity); pumps start and stop by **float switches**; standby pump provided
- Rising mains (force mains): velocity **1.0 – 2.0 m/s**; air valves and scour valves provided; flushing arrangement
- Design capacity = **peak flow** (2 to 3 x avg); station must have **screening, emergency power supply, alarms, ventilation**

12. Drainage in Buildings and Plumbing Systems

Building drainage conveys waste water and soil from fixtures to the public sewer, safely and hygienically. Design follows the **National Building Code and IS 1742 / IS 2065**.

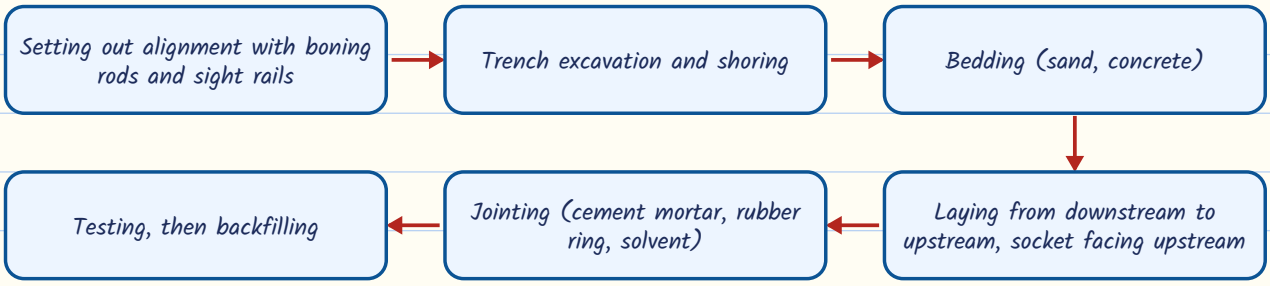
Fixtures and traps

- **Soil pipe** - carries discharge from WCs and urinals; **waste pipe** - from wash basins, baths, sinks, kitchen; **anti-siphonage pipe** (vent) - protects water seals; **vent (stack) pipe** - upto roof, 2 m above roof, **rain water pipes**
- **Traps** - U-shaped bends holding a **water seal** (25-75 mm) to prevent entry of foul gas into the building: P-trap, S-trap, Q-trap, gully trap, intercepting trap (between house drain and sewer), floor trap, grease trap
- **Seal can be lost** by **siphonage, back pressure, evaporation, capillary action** - prevented by venting (anti-siphonage pipes)

System of plumbing	Description
Single stack system	Single pipe for soil and waste; fixtures close to stack; traps with deep seals
One pipe system	Soil and waste in a common stack, **with vents** to each trap
Two pipe system	Separate soil stack and waste stack, with vent pipes; **most hygienic** but costly
Partially ventilated single stack	Reduced vents, for low-rise buildings

- Gradient of house drains 1 in 40 (100 mm) to 1 in 60 (150 mm); **minimum 100 mm diameter**
- **Inspection chambers** at every change; **intercepting trap** at the last chamber; sewage connected to public sewer through a **Y-junction**
- **Rain water pipes** must be separate; **septic tank** in the absence of sewers
- **Testing:** water test (head 1.5 m), smoke test, air test, peppermint test to find leaks

13. Construction, Testing and Maintenance of Sewers



- **Laying** starts at the lower end; the pipes are laid on a **uniform gradient** using a **boning rod and traveller**; trench width = D + 0.3 m each side; **concrete cradle /**

haunch for heavy loads

- **Joints:** cement mortar (1:1), collar, **rubber ring (flexible) joints**, bituminous, lead
- **Testing of sewers: water test** (head of 1.5 m above crown at upstream end, fall in water level checked over 30 min), **air test** (pressure 100 mm water gauge falling by < 25 mm in 5 min), **smoke test**, **mirror and lamp test** for alignment, **ball test** for obstruction
- **Cleaning:** flushing, rodding, **sewer cleaning machines (jetting, bucket machines, hydraulic rodding)**; removal of silt, grease, roots; **record maps** and regular inspection

Sewer gases and safety

- **Hydrogen sulphide** (toxic, 0.1% fatal), **methane** (explosive, 5–15%), **carbon dioxide, carbon monoxide, ammonia**, oxygen deficiency in manholes
- **Safety precautions:** test the atmosphere, **forced ventilation**, no naked flame or smoking, wear **breathing apparatus and safety harness**, never enter a sewer alone, rescue arrangement; **manual scavenging is banned by law**

Example: invert levels of two consecutive sewers

Manhole M1 invert = 100.00 m; sewer M1–M2 length 60 m, gradient 1 in 200, D = 300 mm. Fall = 60/200 = 0.30 m.

Invert of the first sewer at M2 = 99.70 m. If the downstream sewer M2–M3 has D = 450 mm, **match the crown levels**: crown of upstream sewer at M2 = 99.70 + 0.30 = 100.00 m, so the invert of the larger pipe = 100.00 − 0.45 = **99.55 m** (drop of 0.15 m in the manhole).

Remember for exam

* Sewage = 99.9% water; 80% of water supply becomes sewage; per capita BOD 0.08 kg/day
* Min velocity 0.6 m/s, max 3 m/s; max Q at 0.94D, max V at 0.81D
* Rational formula Q = CiA/360; Manning n = 0.013 for sewers
* Corrosion due to H_2S -> H_2SO_4 at crown; control: keep aerobic, ventilation, lining
* Separate system is best for India; manhole spacing 30 m (small sewers)
* Traps maintain water seal to stop sewer gas

Important questions

1. Explain the characteristics and composition of sewage. Define BOD, COD and population equivalent.
2. Compare the separate and combined systems of sewerage.
3. Explain how the quantity of sanitary sewage is estimated.

4. Describe materials used for sewers with their merits and demerits.
5. Explain self-cleansing velocity. Design a circular sewer for a given population.
6. Explain storm runoff estimation by the Rational method.
7. Describe sewer appurtenances (manhole, drop manhole, inverted siphon, catch basin) with sketches.
8. What is sewer corrosion? Explain its mechanism, prevention and control.
9. Explain building drainage, traps and the one-pipe and two-pipe systems of plumbing.